

Teacher and School Recommendations, Asymmetric Information, and Student Choice

A panoramic, annotated reading list — emphasis on empirical / reduced-form work, with key structural and theory anchors flagged.

Project on track recommendations in the Belgian (LOSO) setting.

How to read / annotate this. Organised around four questions: (A) do schools/teachers hold information students and parents lack; (B) why recommendations should be useful (causal / predictive / matching value); (C) how schools form recommendations (determinants, bias, statistical discrimination); (D) how students/parents respond. Sections E–I add the class-rank channel, the belief-formation framework, the statistical-discrimination toolkit, tracking foundations, and reviews, for context. *Tags:* **RF** reduced-form empirical; **EXP** field/lab/survey experiment; **STR** structural; **THY** theory; **REV** review. After each entry there is a commented `% \note{}` line in the source — uncomment it and type your annotation; it renders as a brown indented note. A few venues are flagged (*verify*) where the published outlet/year should be double-checked before citing.

A. Do schools and teachers know more than students and parents?

Evidence that subjective teacher assessments carry information beyond test scores, and that parents/students hold inaccurate beliefs an institution could (or does) move.

Jackson, C. K. (2018). What Do Test Scores Miss? The Importance of Teacher Effects on Non-Test Score Outcomes. *Journal of Political Economy* 126(5).

Teacher effects on behaviours (absences, suspensions, course grades, grade progression) are only weakly correlated with their effects on test scores, yet predict high-school completion and other long-run outcomes more strongly. Core evidence that what teachers observe/affect beyond scores is outcome-relevant — the premise behind a school “information set”. [RF]

Kinsler, J., and Pavan, R. (2016). Parental Beliefs and Investment in Children: The Distortionary Impact of Schools. HCEO Working Paper 2016-029, Univ. of Chicago.

Parents’ beliefs about their child’s skill relative to same-age peers are distorted by the child’s rank *within their own school* (big-fish-little-pond): parents in high-achieving schools underestimate, those in low-achieving schools overestimate, and this feeds investment (homework help, tutoring). The “Kinsler/Pavan” reference — schools shaping the information households act on. [RF / STR]

Dizon-Ross, R. (2019). Parents’ Beliefs about Their Children’s Academic Ability: Implications for Educational Investments. *American Economic Review* 109(8): 2728–65.

Field experiment in Malawi: parents’ baseline beliefs are often wrong and information frictions are worse for the poor. Clear, digestible performance information shifts beliefs and reallocates investment toward better-matched inputs. Clean causal evidence that an institution holding accurate performance information changes household decisions. [EXP]

Boneva, T., and Rauh, C. (2018). Parental Beliefs about Returns to Educational Investments — The Later the Better? *Journal of the European Economic Association* 16(6): 1669–1711.

Elicits parents’ beliefs about the *technology* mapping investments to child outcomes; beliefs vary widely, predict actual investment, and differ by SES (lower-SES parents perceive lower returns). Directly supports modelling SES-graded priors over the skill-to-outcome mapping. [EXP / survey]

Murphy, R., and Wyness, G. (2020). Minority Report: The Impact of Predicted Grades

on University Admissions (and related work on teacher-predicted grades, UK).

UK students apply with teacher-predicted exam grades; predictions are informative but systematically inaccurate and group-skewed (high-achieving low-SES students under-predicted). Useful for the noise/bias structure of a school-issued forecast. Verify exact outlet/year. [RF (verify)]

[Dutch predictive-validity studies] e.g. “The predictive power of track recommendations in Dutch secondary education” and “Does the teacher beat the test? The value of the teacher’s assessment in predicting student ability”.

Dutch micro-data finding the subjective teacher assessment is roughly twice as predictive as primary-school cognitive scores for initial track placement, and more predictive of later allocation than scores. Strong “teachers know more” evidence, but confounds prediction with self-fulfilment. Confirm authors/venue. [RF (verify)]

[German comportment-grades study] Assessing Knowledge or Classroom Behaviour? Evidence of Teachers’ Grading Bias (working paper / EER-type outlet).

Exploits staggered introduction of conduct grades across German states to estimate causal effects on school-to-work transitions and non-cognitive skills, and documents that subject grades themselves encode behaviour. Speaks to the soft (non-test) content teachers fold into evaluations. Verify authorship. [RF (verify)]

B. Why should recommendations be useful? Causal and matching value

Whether teacher expectations/recommendations causally improve (or distort) outcomes, whether discretion adds value as a matching device, and how a teacher mark signals within the system.

Papageorge, N. W., Gershenson, S., and Kang, K. M. (2020). Teacher Expectations Matter. *Review of Economics and Statistics* 102(2): 234–251.

Two teachers reported expectations for the same student; their disagreement (a measurement-error design) identifies a causal effect. Teacher expectations raise the probability of college completion (elasticity ≈ 0.12); teachers are on average over-optimistic, with biases acting like self-fulfilling prophecies. The key “expectations are causal” paper. [RF]

Gershenson, S., Holt, S. B., and Papageorge, N. W. (2016). Who Believes in Me? The Effect of Student–Teacher Demographic Match on Teacher Expectations. *Economics of Education Review* 52.

Two teachers’ differing expectations for the same student reveal systematic demographic-match effects (e.g. non-Black teachers hold lower expectations for Black students). Underpins the idea that recommendations embed the evaluator’s identity/ priors, not just the student’s signal. [RF]

Diamond, R., and Persson, P. (2016). The Long-term Consequences of Teacher Discretion in Grading of High-stakes Tests. NBER Working Paper 22207.

Bunching in Swedish math-test distributions shows teachers “bump up” students who had a bad test day (no gender/immigrant discrimination here). A Wald estimator shows a higher discretionary grade raises later grades, graduation, college entry, and earnings — even though grades don’t raise human capital directly. Interprets grades as a signal *within* the education system, with effort/perceived-ability complementarities. Close to your recommendation-as-signal mechanism. [RF]

Mulhern, C. (2022). Beyond Teachers: Estimating Individual Guidance Counselors’ Effects on Educational Attainment. EdWorkingPaper 22-632 (Annenberg/Brown).

Uses quasi-random assignment of students to counsellors in Massachusetts; counsellors vary substantially in their effect on graduation, college attendance, selectivity, and persistence. Direct evidence

that an advisor holding information/guidance moves attainment. Verify published version. [RF (verify)]

Ichino, A., Mealli, F., and Vivieni, J. (2025). Do Test Scores Help Teachers Give Better Track Advice to Students? A Principal Stratification Analysis. Working paper (arXiv 2511.05128 / CRENoS).

Dutch quasi-randomised cutoff lets teachers revise recommendations upward when scores beat expectations; allowing upgrades raises successful placement in demanding tracks by $\geq 6\%$ while misplacing $\approx 7\%$ of weaker students, with a welfare analysis over short- vs long-term losses. Directly on the value of combining discretion with objective signals. [RF]

C. How do schools form recommendations? Determinants, bias, discrimination

The empirical core on what drives recommendations/grades and whether they are biased by gender, ethnicity, or SES — including blind-vs-non-blind designs built to isolate bias.

Carlana, M. (2019). Implicit Stereotypes: Evidence from Teachers’ Gender Bias. *Quarterly Journal of Economics* 134(3): 1163–1224.

Exploiting near-random assignment of students to teachers with different Gender-Science IAT scores: teacher stereotypes widen the math gender gap and push girls toward less demanding high schools, *following the teacher’s track recommendation*, partly via lowered female self-confidence in math. The model paper for bias working through recommendations and beliefs. [RF]

Carlana, M., La Ferrara, E., and Pinotti, P. (2022). Implicit Stereotypes in Teachers’ Track Recommendations. *AEA Papers and Proceedings* 112: 409–414.

Track-recommendation-specific companion to Carlana (2019): teacher implicit bias maps into the recommendations minority/immigrant students receive. [RF]

Alesina, A., Carlana, M., La Ferrara, E., and Pinotti, P. (2018/2024). Revealing Stereotypes: Evidence from Immigrants in Schools. NBER WP 25333; published version in a top journal (verify outlet/year).

Teachers give lower grades to immigrant-origin students relative to blindly scored tests, consistent with stereotypes; informing teachers of their IAT scores reduces the gap. Connects grading bias to malleable evaluator beliefs. [RF / EXP]

Botelho, F., Madeira, R. A., and Rangel, M. A. (2015). Racial Discrimination in Grading: Evidence from Brazil. *American Economic Journal: Applied Economics* 7(4): 37–52.

Holding blindly scored proficiency and behaviour constant, Black students receive lower teacher-assigned math grades than white classmates; heterogeneity is consistent with statistical discrimination. A clean blind-vs-subjective design. [RF]

Lavy, V. (2008). Do Gender Stereotypes Reduce Girls’ or Boys’ Human Capital Outcomes? Evidence from a Natural Experiment. *Journal of Public Economics* 92(10–11): 2083–2105.

Compares blind (external) and non-blind (teacher) exam scores for the same students to recover grading bias; finds bias against boys in this setting. Foundational quasi-experimental template for detecting evaluator bias. [RF]

Lavy, V., and Sand, E. (2018). On the Origins of Gender Gaps in Human Capital: Short- and Long-Term Consequences of Teachers’ Biases. *Journal of Public Economics* 167:

263–279.

Early-grade teacher grading bias has lasting effects on later course choices and achievement, tracing a long-run channel from biased evaluation to specialization. Directly relevant to STEM/non-STEM sorting. [RF]

Terrier, C. (2020). Boys Lag Behind: How Teachers’ Gender Biases Affect Student Achievement. *Economics of Education Review*.

Within-student blind vs. non-blind comparisons: teacher gender bias in grading affects subsequent achievement, with boys disadvantaged in this context. Complements Lavy on direction and magnitude. [RF]

Burgess, S., and Greaves, E. (2013). Test Scores, Subjective Assessment, and Stereotyping of Ethnic Minorities. *Journal of Labor Economics* 31(3): 535–576.

Compares teachers’ subjective assessments with external test scores by ethnicity, finding systematic gaps consistent with stereotyping in subjective evaluation. Key reference on subjective-vs-objective divergence. [RF]

Hanna, R. N., and Linden, L. L. (2012). Discrimination in Grading. *American Economic Journal: Economic Policy* 4(4).

Field experiment in India randomly assigning child characteristics to identical exams; lower-caste children receive lower scores. Experimental proof that evaluator-side perception moves grades. [EXP]

van Huizen, T., and Jacobs, M. (2024). Teacher Bias or Measurement Error? Working paper (arXiv 2401.04200).

In Dutch administrative data, measurement error in true ability explains $\approx 35\text{--}43\%$ of the conditional SES gap in track recommendations — observational “bias” estimates conflate discrimination with attenuation. Essential methodological caveat for any descriptive of “schools base advice on relative skills”. [RF]

Geven, S. (2025). Tracking Procedures and Criteria and the SES Bias in Teacher Track Recommendations. *American Educational Research Journal* (verify issue).

Survey experiment with Dutch teachers: those interpreting school procedure as weighting a student’s home environment show stronger SES bias in recommendations. Points to institutional procedure as a moderator of bias. [EXP]

[Experimental teacher-discrimination studies] e.g. ethnic/social-class bias in essay evaluation (Germany); track-recommendation vs. grading experiments (Hungary, *European Sociological Review* 2025).

Factorial vignette experiments varying names/SES/gender often find little bias in grading but discrimination in expectations/recommendations, conditional on ambiguity — read as support for situational/statistical accounts over pure taste-based ones. [EXP]

D. How do students and parents respond to recommendations?

Compliance, the effect of making recommendations binding vs. advisory, and SES heterogeneity — mostly reform-based natural experiments.

Bach, M. (2023). Heterogeneous Responses to School Track Choice: Evidence from the Repeal of Binding Track Recommendations. *Economics of Education Review*.

Exploits German states’ repeal of binding recommendations (free choice). Free choice does not widen the SES gap in academic-track enrolment, because previously-eligible low-SES students also move up;

a key mechanism is preferences over the intermediate track. The cleanest design for separating the binding/authority arm from the informational arm. [RF]

Bach, M., and Fischer, M. (2020/2021). Understanding the Response to High-Stakes Incentives in Primary Education / binding recommendations and achievement. ZEW WPs.

Binding-recommendation regimes raise measured achievement but also stress, and parents send children to higher tracks than recommended once recommendations become non-binding. Useful on compliance and parental override. [RF]

Piopiunik, M. (2014). The Effects of Early Tracking on Student Performance: Evidence from a School Reform in Bavaria. *Economics of Education Review* 42: 12–33.

Reform-based evidence on how track placement (in a recommendation-driven system) affects performance. Background on the stakes of the placement recommendations steer. [RF]

[Dutch test-cutoff designs] High-Stakes Testing and Educational Careers (J. Research on Educational Effectiveness, 2023); “The quality of school track assignment decisions by teachers” (working paper, arXiv 2304.10636).

Use discontinuities in test-based recommendation cutoffs (and differences across test providers) to instrument for first-year track, estimating how marginal recommendation upgrades change educational careers. Direct evidence on compliance at the margin. [RF]

E. Ordinal rank and relative-standing signals (the class-rank channel)

Directly relevant because your school signal Z is class-relative rank in grades/effort: how relative-position signals move confidence, beliefs, and subject choice independently of absolute ability.

Murphy, R., and Weinhardt, F. (2020). Top of the Class: The Importance of Ordinal Rank. *Review of Economic Studies* 87(6): 2777–2826.

Using the universe of English students and variation in achievement distributions across primary classes, ordinal rank has lasting effects on secondary-school test scores, confidence, and subject choice — independent of underlying ability, and even though later peers/teachers don’t know the prior rank. Effects are larger for boys and feed the math-course gender gap. Central to the class-rank mechanism. [RF]

Elsner, B., and Isphording, I. E. (2017). A Big Fish in a Small Pond: Ability Rank and Human Capital Investment. *Journal of Labor Economics* 35(3): 787–828.

Comparing cohorts within the same school, a higher ordinal rank (holding ability fixed) raises later educational attainment and expectations. Reinforces that relative standing, not just absolute skill, shapes choices. [RF]

Denning, J. T., Murphy, R., and Weinhardt, F. (2023). Class Rank and Long-Run Outcomes. *American Economic Journal: Applied Economics* (NBER WP 27468; verify issue).

Extends the rank literature to long-run outcomes (earnings, attainment), confirming that early relative standing has persistent consequences. [RF]

Goulas, S., and Megalokonomou, R. (2021). Knowing Who You Actually Are: The Effect of Feedback on Short- and Longer-Term Outcomes. *Journal of Economic Behavior &*

Organization.

Studies how giving students feedback on their relative performance (rank information) changes effort and subsequent outcomes. A direct empirical analog of disclosing the kind of relative-standing information embedded in Z — relevant to your CF1. [RF]

F. Belief formation and information frictions in educational choice

The framework for how a signal (grade, recommendation, disclosed information) shifts beliefs and choices — awareness, ambiguity, learning about comparative advantage, information experiments. Some are structural; flagged.

Giustinelli, P., and Pavoni, N. (2017). The Evolution of Awareness and Belief Ambiguity in the Process of High School Track Choice. *Review of Economic Dynamics* 25: 93–120.

Repeated survey of Italian 8th-graders: students are only partially aware of available tracks and hold ambiguous beliefs about completing each, with disadvantaged students acquiring information more slowly. Direct evidence on the belief object a recommendation acts on, and on SES heterogeneity in updating. [RF / survey]

Giustinelli, P. (2016). Group Decision Making with Uncertain Outcomes: Unpacking Child–Parent Choice of the High School Track. *International Economic Review*.

Models and estimates the joint child–parent track decision using elicited subjective expectations, quantifying how information/sensitization campaigns would shift the track distribution. Relevant if parental influence matters in your setting. [STR / RF]

Wiswall, M., and Zafar, B. (2015). Determinants of College Major Choice: Identification Using an Information Experiment. *Review of Economic Studies* 82(2): 791–824.

Provides students true population information on major-specific outcomes, generating experimental belief variation; students revise logically and choices respond, though tastes dominate. The template for treating an external signal as a belief shifter about returns. [EXP]

Wiswall, M., and Zafar, B. (2015). How Do College Students Respond to Public Information about Earnings? *Journal of Human Capital* 9(2).

Companion: most students are non-Bayesian updaters, and assuming Bayesian processing substantially overstates the welfare gains from information. Important caution for any moment-matching/Bayesian-updating assumption about how students absorb advice. [EXP]

Hastings, J., Neilson, C., and Zimmerman, S. (2015). The Effects of Earnings Disclosure on College Enrollment Decisions. NBER WP; and Hastings, Neilson, and Zimmerman (2013) on returns to majors.

Field experiment disclosing program-level earnings: muted average enrolment effects, but low-SES applicants shift toward higher-net-return degrees. Real-world analog of an informed party moving disadvantaged students' choices. [EXP / RF]

Li, H., and Xia, X. (2024). Grades as Signals of Comparative Advantage: How Letter Grades Affect Major Choices. *Journal of Economic Behavior & Organization* 227: 106717.

Regression discontinuity at letter-grade cutoffs in intro courses: students just above a cutoff are more likely to major in that field — they learn comparative advantage from a noisy grade signal. Direct causal evidence for the signal-to-belief-to-choice chain. [RF]

Aucejo, E., and James, J. (2021). The Path to College Education: The Role of Math and Verbal Skills. *Journal of Political Economy* 129(10): 2905–2946.

Shows relative (math vs. verbal) comparative advantage shapes the path into college and field, beyond absolute skill. Useful microfoundation for STEM/non-STEM sorting on comparative advantage — the dimension your subtrack recommendation encodes. [STR / RF]

Kinsler, J., and Pavan, R. (2015). The Specificity of General Human Capital: Evidence from College Major Choice. *Journal of Labor Economics* 33(4): 933–972.

Structural model of major choice with skill uncertainty and comparative advantage; quantifies mismatch costs. Background for modelling STEM/non-STEM match quality and learning. [STR]

Stinebrickner, T., and Stinebrickner, R. (2012; 2014). Learning about Academic Ability and the College Dropout Decision / Academic Performance and College Dropout. *Journal of Labor Economics* 30(4) and 32(3): 601–644.

Foundational learning models using longitudinal expectations data: students update Bayesianly about ability from grades and revise enrolment. The canonical “learning about own ability from signals” framework the recommendation extends. [STR]

Arcidiacono, P., Aucejo, E., Maurel, A., and Ransom, T. (2025). College Attrition and the Dynamics of Information Revelation. *Journal of Political Economy* 133(1): 53–110.

State-of-the-art dynamic model in which students learn about match quality from signals while choosing majors and attrition. The structural backbone many recommendation models build on (signals arrive exogenously here — the gap a strategic-school model fills). [STR]

G. Statistical discrimination and ability revelation: interpretive tools

Mostly theory/experiment plus labor-market evidence, included because they give precise language for “is the recommendation accurate, inaccurate, or taste-based?” and for how discrimination fades as ability is revealed.

Bohren, J. A., Haggag, K., Imas, A., and Pope, D. G. (2025). Inaccurate Statistical Discrimination: An Identification Problem. *Review of Economics and Statistics* 107(3): 605–620.

Inaccurate beliefs are a third source of discrimination beyond taste-based and accurate statistical; when unaccounted for, they get misclassified as taste-based. Proposes three identification routes — varying signals, eliciting beliefs, correcting beliefs — all with analogs in rich tracking data with subjective-belief measures. The linchpin for separating your channels. [THY / EXP]

Bohren, J. A., Imas, A., and Rosenberg, M. (2019). The Dynamics of Discrimination: Theory and Evidence. *American Economic Review* 109(10).

Models and tests how discrimination evolves as evaluators learn, with feedback that can entrench gaps. Useful for the self-fulfilling channel when recommendations affect later skill accumulation. [THY / EXP]

Bordalo, P., Coffman, K., Gennaioli, N., and Shleifer, A. (2016). Stereotypes. *Quarterly Journal of Economics* 131(4): 1753–1794.

Representativeness-based model of beliefs: group stereotypes carry a “kernel of truth” but are exaggerated. A disciplined way to parametrize how a school’s group-conditioned beliefs depart from accuracy. [THY / EXP]

Altonji, J. G., and Pierret, C. R. (2001). Employer Learning and Statistical Discrimination. *Quarterly Journal of Economics* 116(1): 313–350.

Canonical employer-learning model and test: when firms statistically discriminate on an easily observed trait (e.g. education) early on, its wage weight falls as harder-to-observe productivity is revealed. The template for “signals matter more when true ability is hidden” — relevant to why recommendations carry weight early in the tracking process. [RF / THY]

Arcidiacono, P., Bayer, P., and Hizmo, A. (2010). Beyond Signaling and Human Capital: Education and the Revelation of Ability. *American Economic Journal: Applied Economics* 2(4): 76–104.

Ability is revealed quickly for college graduates but slowly for high-school workers, so employers statistically discriminate (e.g. by race) in the high-school market but not the college market. A concrete instance of an institution’s signal substituting for, and then being displaced by, revealed ability. [RF / STR]

Phelps, E. (1972), *AER*; Arrow, K. (1973), “The Theory of Discrimination”; Aigner, D., and Cain, G. (1977), *ILR Review* 30(2).

Classical statistical-discrimination foundations: group averages as the optimal response to noisy individual signals (Phelps/Aigner–Cain) and self-fulfilling equilibria (Arrow). Background for why conditioning a recommendation on group can be “rational” yet costly. [THY]

H. Tracking: foundational reduced-form context

The broader tracking literature situating why placement matters and what is at stake in the recommendation.

Card, D., and Giuliano, L. (2016). Can Tracking Raise the Test Scores of High-Ability Minority Students? *American Economic Review* 106(10): 2783–2816.

Ability tracking/placement can raise outcomes for high-ability minority students, highlighting that who is placed where (and how it is decided) has real consequences. [RF]

Dustmann, C., Puhani, P. A., and Schönberg, U. (2017). The Long-Term Effects of Early Track Choice. *The Economic Journal* 127.

Estimates long-run labour-market and attainment effects of early between-school tracking, with attention to track mobility. Motivates the stakes and irreversibility of the age-14 decision. [RF]

Hanushek, E. A., and Woessmann, L. (2006). Does Educational Tracking Affect Performance and Inequality? Differences-in-Differences Evidence Across Countries. *The Economic Journal* 116(510): C63–C76.

Cross-country diff-in-diff: early tracking tends to increase inequality without clear average gains. Canonical macro evidence on tracking and inequality. [RF]

De Groote, O. (2025). Dynamic Effort Choice in High School: Costs and Benefits of an Academic Track. *Journal of Labor Economics* 43(2): 467–502.

Structural treatment of the same Belgian/LOSO setting (effort, track, performance dynamics). The natural in-house anchor for data and institutional detail. [STR]

Betts, J. R. (2011). The Economics of Tracking in Education. In *Handbook of the Economics of Education*, vol. 3: 341–381.

Survey of tracking designs, identification challenges, and findings. Good orienting reference for the tracking literature as a whole. [REV]

I. Reviews and surveys

Geven, S., and co-authors (2022). Are Tracking Recommendations Biased? A Review of Teachers' Role in the Creation of Inequalities in Tracking Decisions. *Teaching and Teacher Education*.

Systematic review of ~ 27 studies: recommendations are biased against low-SES students; ethnic-bias evidence is mixed; observed characteristics don't account for the bias, pointing to institutional moderators. Best single entry point to the bias literature. [REV]

Zanga and De Gioannis (2023). Discrimination in Grading: A Scoping Review. *Studies in Educational Evaluation*.

Catalogues 37 grading-discrimination studies by method (experiments; blind-vs-non-blind) and by characteristic (gender, race/ethnicity, migration, SES). Handy map of designs and findings. [REV]

Wenz, S., and Hoenig, K. (2020). Ethnic and Social Class Discrimination in Education: Experimental Evidence (review/overview; verify outlet).

Frequently-cited overview distinguishing grading vs. expectation/recommendation bias and the conditions under which each appears. Verify exact venue/year. [REV (verify)]

Notes on use. Most on-theme empirical anchors: schools-know-more — Jackson (2018), Kinsler–Pavan (2016), Dizon-Ross (2019), Boneva–Rauh (2018); recommendations as causal — Papageorge–Gershenson–Kang (2020), Diamond–Persson (2016), Mulhern (2022); the class-rank channel for Z — Murphy–Weinhardt (2020), Elsner–Isphording (2017); bias/discrimination in how schools decide — Carlana (2019) and the blind-vs-non-blind designs (Botelho et al. 2015; Lavy 2008; Lavy–Sand 2018); how students respond — Bach (2023) and the Dutch cutoff designs; interpretive toolkit — Bohren et al. (2025), Altonji–Pierret (2001), Arcidiacono–Bayer–Hizmo (2010). Items tagged STR/THY are for panorama and framing rather than reduced-form templates. Confirm any venue marked (*verify*) before citing.